Toward a four-dimensional Yang–Mills construction

Notebook 19: the physical background Gaussian in the source coordinates

Research working document

17 October 2026

Abstract

We construct the actual homogeneous current-step pivot chart near the trivial field and a physical stationary background for the next ordinary averaging constraint. The chart is local and leaves the retained coordinates unchanged. Two minimum comparisons use the previously constructed ordinary minimizer to identify the current classical action on the next fiber. An explicit rooted gauge sets the source's averaged contour variable to the identity. The resulting reduced fluctuation Hessian is a small perturbation of the flat retained precision, uniformly in earlier averaging depth and periodic volume. Its formula keeps the nonlinear constraint curvature and the conversion from relative exponential to ordinary potential coordinates. This gives a uniform complex covariance neighborhood for the physical small-background branch. We also isolate the local measure and gauge factors needed by the interacting integral.

1 From a classical minimum to the source Gaussian

The existence argument needs the Gaussian in the actual quantum integration coordinates. The fine constrained inverse and the Hessian of a minimized coarse action are useful inputs, but neither alone identifies that Gaussian. The source also changes coordinates, eliminates pivot bonds, and includes an exponential gauge term. Away from the flat field these operations contribute current-dependent second derivatives.

Here the current lattice has unit spacing, the previous fine spacing is M^{-1}, and the next spacing is a fixed odd L. The background is in a sufficiently small global potential ball. The constants are uniform in $M = L^k$ and compatible periodic volume, with the specified source block/interface geometry. The result concerns physical stationary backgrounds and their analytic extension near the trivial field. The original larger orbit domain, freely prescribed independent currents, and mixed large-field histories require further estimates.

2 The physical current-step chart and Gaussian precision

The current lattice has spacing one, the original lattice spacing is M^{-1} with $M = L^k$, and the next spacing is fixed odd L. We work on compatible periodic tori with the ordinary block and corridor geometry. The original CMP109 choice has $L > 11$; the local arguments below work for odd $L \geq 3$. Positive current bonds and real color coordinates orthonormal for the normalized trace are counted once. Current pairings use counting measure. Transposes are bilinear after complexification; numerical-range bounds use the Hermitian pairing.

Let $S_M(a)$ be the coupling-free Wilson action at the ordinary homogeneous minimizer with current datum $V = e^{ia}$. Notebooks 15–16 give, on a uniform complex supremum ball,

$$m_M(a) = DS_M(a), \qquad \|m_M(a)\|_\infty \leq C \|a\|_\infty , \qquad \left\| D^2 S_M(a) - q_M \right\|_\kappa \leq C_\kappa \|a\|_\infty . \tag{1}$$

Here q_M is the physical flat Maxwell precision. The norm $\|\cdot\|_\kappa$ is the maximum of the exponentially weighted absolute row and column sums, with a fixed finite number of color and orientation coordinates per cell. It is a matrix algebra norm. We choose κ below the common exponent of Notebooks 07 and 16. Regrouping unit cells into fixed L-blocks only changes fixed constants and the exponent.

Let K_L insert the pivot variables in the flat rooted constraint, and let T_L be the averaged root-to-site contour derivative. Notebook 17 proves

$$\begin{aligned} A_{M,0} = K_L^T(q_M + T_L^T T_L)K_L, \qquad \gamma_* I \le A_{M,0} \le B_* I, \\ \|A_{M,0}\|_\kappa + \left\|A_{M,0}^{-1}\right\|_\kappa \le C_*. \end{aligned} \tag{2}$$

Constants below may depend on L, dimension, the color normalization and κ, but not on M or periodic volume.

2.1 The actual ordinary average and its relative coordinates

The source formulas are CMP109, pp. 253–255 and 265–268, (0.11)–(0.16), (2.3)–(2.13). Raw I, lines 46952–47000, and Raw II, lines 9045–9095 and 73865–73945, spell out their normalization. The following argument uses these explicit local maps, not a general-background estimate quoted from the deposit.

Write $P_V(y,x)$ for the analytic group mean of the prescribed shortest root-to-x contour products in the block centered at y. The mean is analytic near identity, its first derivative there is the arithmetic mean, and it is equivariant at both endpoints. With $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$,

$$P_{V^u}(y,x) = u(y)P_V(y,x)u(x)^{-1}. \tag{3}$$

All these contours are block-internal. For $c = (y, y + Le_\mu)$, $x' = x + Le_\mu$, and the straight reference transporter $R_c(V) = V([y, y + Le_\mu])$, the ordinary average is

$$\begin{aligned} U_x(V) &= P_V(y,x)V([x,x'])P_V(y+Le_\mu,x')^{-1}, \\ \mathcal{M}_L(V)(c) &= \exp\left\{L^{-d}\sum_{x\in B(y)} \log[U_x(V)R_c(V)^{-1}]\right\} R_c(V). \end{aligned} \tag{4}$$

This is the ordinary rooted map, not the fixed-reference corrected average. It reads two endpoint blocks and their connecting strings, and obeys

$$\mathcal{M}_L(V^u) = \mathcal{M}_L(V)^{u|_{\text{block centers}}}.$$

Define the source's left-relative coordinates

$$\begin{aligned} F_V(B;c) &= \frac{1}{i}\log[\mathcal{M}_L(e^{iB}V)(c)\mathcal{M}_L(V)(c)^{-1}], \\ G_V^{\mathrm{rel}}(B;y,x) &= \frac{1}{i}\log[P_{e^{iB}V}(y,x)P_V(y,x)^{-1}], \\ A_V &= D_B F_V(0) = LQ_V^\flat, \qquad T_V = D_B G_V^{\mathrm{rel}}(0). \end{aligned} \tag{5}$$

A_V is a rectangular constraint derivative; the reduced precision will be written $A_M(V)$. The source kernel convention is

$$(Q_V^\flat B)(c) = \sum_b L^{-d} Q_V^\flat(c,b)B(b). \tag{6}$$

Lemma 2.1 (One pivot per nonlinear equation)**.** *Let $p(c)$ be the middle current bond on the line through the centers of c, crossing their common interface. The full nonlinear equation $F_V(B;c)$ reads $p(c)$ and no other pivot. For $a = \log V/i$ sufficiently small,*

$$\pi_c(V) = Q_V^\flat(c, p(c)) = I + O(\|a\|_\infty), \qquad h_V(c) = L^{d-1}\pi_c(V)^{-1}, \qquad A_V h_V = I. \tag{7}$$

Here h_V inserts only at the corresponding pivots. The linear insertion K_V retains every nonpivot coordinate and sets

$$(K_V z)(p(c)) = -\pi_c(V)^{-1} \sum_{b \ nonpivot} Q_V^\flat(c, b) z(b). \tag{8}$$

Thus $A_V K_V = 0$ and $K_1 = K_L$.

Proof. No internal root contour contains a pivot. A string for another coarse bond has a different orientation or a different pair of endpoint blocks, and cannot contain $p(c)$. This proves the nonlinear incidence assertion. At identity, precisely L starting points in the straight-string average contain $p(c)$: their transverse coordinates are fixed and their longitudinal coordinate runs through the first block. Its coefficient in $Q_L^\flat$ is therefore L^{-d}, and in A_1 it is L^{1-d}. Analyticity gives the displayed inverse on a fixed small neighborhood. The common factor L^{1-d} cancels when solving the linear equation, proving (8). □

2.2 Exact straightening, finite range, and the cutoff

Write $F_V(B) = A_V B + C_V(B)$, where C_V starts at degree two. The source straightening map is the solution of

$$\widetilde{D}_V(B) = C_V(B - h_V \widetilde{D}_V(B)). \tag{9}$$

For each output c, the right side depends on $\widetilde{D}_V(c)$ alone, by Lemma 2.1. A finite-dimensional contraction, with derivative zero at $B = 0$, solves this equation uniquely and analytically on a uniform small ball. There are finitely many local cell types. No iteration propagates support to another constraint equation. Set

$$\Phi_V(z) = K_V z - h_V \widetilde{D}_V(K_V z). \tag{10}$$

Then

$$F_V(\Phi_V(z)) = 0, \qquad \Phi_V(z)(b) = z(b) \quad (b \text{ nonpivot}), \tag{11}$$

and this is the unique small solution for the specified retained coordinates. It is therefore the actual source chart, rather than a different parametrization of the same fiber. Its first two derivatives are

$$\begin{aligned} D\Phi_V(0) &= K_V, \\ D^2\Phi_V(0)[u, v] &= -h_V D^2 F_V(0)[K_V u, K_V v] = -2h_V C_V^{(2)}(K_V u, K_V v). \end{aligned} \tag{12}$$

$C_V^{(2)}$ denotes the bilinear homogeneous coefficient whose diagonal is the quadratic term, not the full Hessian.

For precision estimates it is useful to record how finite dimensional analyticity gives uniform weighted bounds. Let N_L bound the number of scalar coordinates read by a local output, I_L the number of output neighborhoods containing a fixed input coordinate, and ℓ_L their diameter. Cauchy on a fixed smaller local polydisc bounds coordinate derivatives through order three by C_L. In particular, for $b = \|a\|_\infty$,

$$\begin{aligned} \|K_V\|_\kappa + \|h_V\|_\kappa + \|T_V\|_\kappa &\le C_{L,\kappa}, \\ \|K_V - K_L\|_\kappa + \|h_V - h_1\|_\kappa + \|T_V - T_L\|_\kappa &\le C_{L,\kappa} b. \end{aligned} \tag{13}$$

The first and second fluctuation derivatives of Φ_V, and its first and second background derivatives, have fixed finite range and uniform coefficient sums on a smaller ball. The corresponding second fluctuation derivative differs from its identity-background value by $O(b)$, by integrating its bounded background derivative. The same argument covers the mixed derivatives of total order at most three.

More explicitly, if a local map f_r has second coordinate derivatives bounded by C_L, define the contracted matrix by $\langle u, H_s v\rangle = \sum_r s_r D^2 f_r[u,v]$. At most I_L outputs contain a fixed input, and each has at most N_L other inputs. Hence

$$\|H_s\|_\kappa \le C_L I_L N_L e^{2\kappa\ell_L} \|s\|_\infty . \tag{14}$$

This proves the required row and column bounds without a volume factor. The first-derivative estimates use the same incidence count.

After the source coupling scaling, the original relative link field is $\Phi_V(gz)$. Exactly,

$$\Phi_V(gz)(b) = gz(b) \quad (b \text{ nonpivot}). \tag{15}$$

Thus a retained-link product of indicators $|B'(b)| < r$ becomes $|z(b)| < r/|g|$, with no changed coordinate argument. CMP109, p. 267, explains the resulting control of eliminated pivots; CMP116, (2.2)–(2.3), explicitly uses the retained bond set. Raw II's abbreviated description as a product over bonds must not be used to discard other indicators. Any additional pivot, contour or reconstructed-field characteristic function remains at its exact pullback through $\Phi_V(gz)$. No equality between such an enlarged window and a rectangular retained window is claimed.

2.3 The exponential gauge factor

With normalized trace $\tau(I) = 1$, the analytic absolute gauge function is

$$\mathcal{G}(V) = \sum_y \sum_{\substack{x\in B(y)\\ x\neq y}} \left\{ 1 - \frac{1}{2}\tau[P_V(y,x) + P_V(y,x)^{-1}] \right\}. \tag{16}$$

On real unitary fields this is the source sum of $1 - \Re\tau P_V$. Its step factor is $\exp[-g^{-2}\mathcal{G}(V)]$; field-independent normalization and source contour indicators stay separate.

Suppose $P_V(y,x) = 1$ for every pair. Since $1 - \frac{1}{2}\tau(e^{iX} + e^{-iX}) = \frac{1}{2}\tau(X^2) + O(X^4)$,

$$\begin{aligned} &\mathcal{G}(V) = 0, \qquad D_B\mathcal{G}(e^{iB}V)|_{B=0} = 0,\\ &D_z^2\mathcal{G}(e^{i\Phi_V(z)}V)|_{z=0} = K_V^T T_V^T T_V K_V. \end{aligned} \tag{17}$$

The nonlinear contour term does not contribute at quadratic order because the first derivative of the penalty vanishes. This is the averaged-contour penalty, not a tree penalty.

There is also an exact simplification. Contours are internal to blocks and contain no pivots. If E_f extends retained coordinates by zero on pivots, then

$$\mathcal{G}(e^{i\Phi_V(z)}V) = \mathcal{G}(e^{iE_f z}V), \qquad T_V h_V = 0, \qquad T_V K_V = T_V E_f. \tag{18}$$

The gauge factor is thus literally a local function of the unchanged retained coordinates. Its cubic remainder after the quadratic term is bounded by C_L times the sum of cubed local suprema of $|z|$. After $z \mapsto gz$ and division by g^2, this is $O(g)$ times that local cubic sum, as long as $|g|\,\|z\|_\infty$ stays in the fixed chart radius.

At a general background with $P_V \neq 1$, the absolute function (16) has a nonzero gradient and a contour-curvature Hessian residual. These terms are retained in its analytic extension and are $O(\|a\|_\infty)$ perturbations of the flat coefficients. The formal gradient times $D^2\Phi_V$ vanishes by (18). Replacing the absolute penalty by a relative one away from $P_V = 1$ would change the source function and is not done here.

2.4 The physical stationary background

Proposition 2.2 (Stationary homogeneous branch in the source gauge)**.** *For $W = e^{iw}$ with $\|w\|_\infty$ sufficiently small, there is an analytic current configuration $V_*(W)$ satisfying*

$$\mathcal{M}_L(V_*) = W, \qquad P_{V_*}(y,x) = 1, \qquad \|\log V_*/i\|_\infty \le C_L \|w\|_\infty . \tag{19}$$

For real data it is a constrained local minimum of the classical action, and a stationary point of the action plus the absolute exponential-gauge function.

Proof. Apply Notebook15 with final spacing L, original spacing $1/M$ and total ratio ML. Under $x' = x/L$, this is the already established unit-final-spacing theorem. In dimension four the coupling-free Wilson action has no extra scale factor; physical potential and derivative norms change only by fixed powers of L. Let $U_f(W)$ be that ordinary minimizer and set

$$V_p(W) = \mathcal{M}^k(U_f(W)).$$

Exact composition gives $\mathcal{M}_L(V_p) = W$. Notebook14's ordinary trajectory and Notebook15's potential bound give $\|\log V_p/i\|_\infty \le C_L \|w\|_\infty$ and a fine physical N_α bound $C_L \|w\|_\infty$.

Let $U_c(V)$ be the current-scale ordinary minimizer. Choose a common current comparison radius δ and then the data radius so that $U_f(W)$ is in the current minimum ball, every $U_c(V)$ with $\|\log V/i\|_\infty < \delta$ is in the final-spacing-L minimum ball, and $\|\log V_p/i\|_\infty < \delta/2$. These are finitely many conditions $C_L\delta < r_0$ and $C_L \|w\|_\infty < \delta/2$, independent of M and volume. Current minimality gives

$$S_M(\log V_p/i) \le \mathcal{S}(U_f(W)).$$

The reverse inequality follows because $U_c(V_p)$ is a full feasible competitor by ordinary composition and lies in its common minimum ball. Hence equality holds. For any V in the current comparison ball with $\mathcal{M}_L(V) = W$, the same full-minimum comparison gives

$$S_M(\log V/i) \ge \mathcal{S}(U_f(W)) = S_M(\log V_p/i). \tag{20}$$

This proves the required constrained minimum directly.

Define a rooted current-site gauge by $u(y) = 1$ at centers and $u(x) = P_{V_p}(y,x)$ in $B(y)$. It is local analytic and $\|\log u/i\|_\infty \le C_L \|\log V_p/i\|_\infty$. Set $V_* = V_p^u$. Equations (3) and (4) give $P_{V_*} = 1$ and $\mathcal{M}_L(V_*) = W$ exactly. The local coarse gauge invariance of Notebook16, with its smooth fine extension and stricter common margins, gives $S_M(\log V_*/i) = S_M(\log V_p/i)$. Therefore (20) applies with V_* as well. The gauge function has zero gradient there by (17).

All the maps are analytic on a uniform complex ball. The constraints and stationarity identities extend from the real form by holomorphic uniqueness. No complex minimum interpretation is asserted. □

2.5 Both chart-curvature contributions to the Hessian

Let $a = \log V/i$ and define the ordinary potential in the source relative chart by

$$\Xi_a(B) = \frac{1}{i}\log(e^{iB}e^{ia}), \qquad a_V(z) = \Xi_a(\Phi_V(z)). \tag{21}$$

For the bondwise left Jacobian $J_a X = \int_0^1 \mathrm{Ad}(e^{ita})X\,dt$,

$$D\Xi_a(0) = J_a^{-1}, \quad J_a^{-1} = I + O(\|a\|_\infty), \quad D^2\Xi_a(0) = O(\|a\|_\infty). \tag{22}$$

The last bound follows from analyticity and $\Xi_0(B) = B$. Put $K_a = J_a^{-1} K_V$. Then

$$\begin{aligned} Da_V(0) &= K_a, \\ D^2 a_V(0)[u,v] &= J_a^{-1} D^2 \Phi_V(0)[u,v] + D^2 \Xi_a(0)[K_V u, K_V v]. \end{aligned} \tag{23}$$

Thus, at a gauge-zero physical background $V_*(W)$, the exact retained-coordinate Hessian of action plus gauge is

$$\begin{aligned} A_M(V) &= K_a^T H_M(a) K_a + \mathrm{Curv}_M(V) + K_V^T T_V^T T_V K_V, \\ \langle u, \mathrm{Curv}_M(V) v \rangle &= \langle m_M(a), D^2 a_V(0)[u,v] \rangle, \qquad H_M(a) = D^2 S_M(a). \end{aligned} \tag{24}$$

The two terms in (23) are respectively the nonlinear constraint curvature and the conversion from relative exponential to ordinary log coordinates. Both occur in (24).

For an explicit check against the source current vertex, set

$$j_{\mathrm{rel}} = J_a^{-T} m_M(a), \qquad H_{\mathrm{left}} = J_a^{-T} H_M(a) J_a^{-1} + \langle m_M(a), D^2 \Xi_a(0) \rangle.$$

Then the same precision is

$$\begin{aligned} \langle u, A_M(V) v \rangle = {} & \langle K_V u, H_{\mathrm{left}} K_V v \rangle + \langle T_V K_V u, T_V K_V v \rangle \\ & - 2 \langle j_{\mathrm{rel}}, h_V C_V^{(2)}(K_V u, K_V v) \rangle. \end{aligned} \tag{25}$$

This has the sign and factor of Raw I's source vertex. Differentiating the same action through a fine minimizing extension writes the physical current as $H_1^T J$; pure gauge pieces pair to zero by gauge invariance. No independent J is introduced here. Constrained stationarity gives

$$K_V^T j_{\mathrm{rel}} = 0, \qquad j_{\mathrm{rel}} = A_V^T \lambda, \quad \lambda = h_V^T j_{\mathrm{rel}}. \tag{26}$$

The second identity follows because $I - h_V A_V$ has range $\ker A_V$. It does not make j_{rel} zero and does not eliminate its contraction with $h_V C_V^{(2)}$.

2.6 Uniform precision, covariance, and uniqueness bounds

Theorem 2.3 (Physical-background retained precision)**.** *For the background of Proposition 2.2, with* $b = \|\log V/i\|_\infty$ *sufficiently small,*

$$\|A_M(V) - A_{M,0}\|_\kappa \le C_{L,\kappa} b. \tag{27}$$

The full Hessian defined using the actual absolute gauge function has the same analytic estimate on a full small current-background ball. At arbitrary backgrounds it need not be centered, since its first derivative can be nonzero. For

$$C_{L,\kappa} b \le \gamma_*/2, \qquad C_* C_{L,\kappa} b \le 1/2, \tag{28}$$

one has

$$\begin{aligned} \mathrm{Re}\langle z, A_M(V) z \rangle_{\mathrm{Herm}} \ge \tfrac{1}{2}\gamma_* \|z\|_2^2, \qquad & \left\| A_M(V)^{-1} \right\|_\kappa \le 2C_*, \\ \left\| A_M(V)^{-1} - A_{M,0}^{-1} \right\|_\kappa \le 2C_*^2 C_{L,\kappa} b. & \end{aligned} \tag{29}$$

The constants are uniform in all preceding classical depths $M = L^k$ *and compatible periodic volumes.*

Proof. Equations (13) and (22) give $\|K_a - K_L\|_\kappa \le C_{L,\kappa} b$, with all unperturbed factors uniformly bounded. Apply the weighted algebra inequality and (1) to the first term in (24). Its difference from $K_L^T q_M K_L$ is $O(b)$. Apply (14) to (23), using $\|m_M(a)\|_\infty \le Cb$, to obtain $\|\mathrm{Curv}_M(V)\|_\kappa \le C_{L,\kappa} b$. The ordinary-log conversion part is even $O(b^2)$, but that improvement is unnecessary. The gauge term differs from its flat value by $O(b)$ through (13). The absolute gauge residual off $P_V = 1$ is a local analytic $O(b)$ term, so the same bound holds for that extension.

The Schur bound controls the Hilbert operator norm. Together with (2), it proves the accretivity assertion. The other bounds follow by the Neumann series in the weighted matrix algebra. □

For clarity, uniqueness of the source critical configuration can be obtained on a uniform common coordinate neighborhood, not merely from the Hessian at a point. Keep one fixed chart Φ_V and let z vary. Local Cauchy gives $D\Phi_V(z) = K_L + O(b + \|z\|_\infty)$, uniformly bounded second derivatives, and $\|a_V(z)\|_\infty \le C_L(b + \|z\|_\infty)$. The same chain-rule proof, including the absolute gauge contour residual, gives

$$\left\| D_z^2 \{ S_M(a_V(z)) + \mathcal{G}(e^{i\Phi_V(z)} V) \} - A_{M,0} \right\|_\kappa \le C_{L,\kappa}(b + \|z\|_\infty). \tag{30}$$

On a smaller real convex supremum ball this Hessian is bounded below by $\gamma_*/2$. There is at most one critical point in that ball. Every sufficiently small configuration on the same fiber belongs to this chart, by uniqueness for its retained relative coordinates. Thus the critical orbit and gauge-zero selection in CMP109 agree with the constructed branch whenever both are in this common homogeneous neighborhood. No global minimum outside that neighborhood is inferred.

For any specified $\varepsilon > 0$, the additional restriction $b \le \varepsilon \gamma_* / C_{L,\kappa}$ gives

$$\left\| A_{M,0}^{-1/2} [A_M(V) - A_{M,0}] A_{M,0}^{-1/2} \right\|_{2\to 2} \le \varepsilon. \tag{31}$$

The counting covariance in the current retained coordinates is $A_M(V)^{-1}$; no factor M^4 is inserted.

Corollary 2.4 (The tunable covariance neighborhood)**.** *Let $C_0 = A_{M,0}^{-1}$, so its real spectral bounds are $\lambda_{\mathrm{cov}} = B_*^{-1}$ and $\Lambda_{\mathrm{cov}} = \gamma_*^{-1}$. Fix any $0 < \theta \le 1/4$ before the coupling is chosen. If, in addition to (28), the background radius satisfies*

$$2C_*^2 C_{L,\kappa} b \le \frac{\theta}{2B_*}, \tag{32}$$

then

$$\left\| A_M(V)^{-1} - C_0 \right\|_{2\to 2} \le \theta \lambda_{\mathrm{cov}} / 2. \tag{33}$$

This is exactly the covariance-neighborhood hypothesis of Notebook18's tunable marked theorem. A common covariance majorant and exponentially weighted row sum are uniform in M and volume.

Proof. Use (29) and the Schur operator bound. The same weighted inverse estimate gives the entry majorant $J_{ij} = 2C_* e^{-\kappa d(i,j)}$. For every smaller exponent $0 < \kappa' < \kappa$, its weighted row sum is uniformly finite by the polynomial growth of the lattice, including periodic distance on compatible tori. The real baseline has the stated spectral bounds by (2). □

The local interaction hypotheses of that marked theorem are still separate. Its strict-background-locality version further requires a support-valued precision expansion and an appropriate product background domain, which the present kernel estimate does not imply. Alternatively, (31) gives relative covariance error at most $\varepsilon/(1-\varepsilon)$; the direct condition (32) matches the stated theorem without that conversion.

At the centered physical background, the exact classical exponent after scaling is

$$\begin{aligned} &- g^{-2} \{ S_M(a_V(gz)) + \mathcal{G}(e^{i\Phi_V(gz)} V) \} \\ &\qquad = -g^{-2} S_M(a) - \tfrac{1}{2} \langle z, A_M(V) z \rangle + \text{terms of positive order in } g, \end{aligned} \tag{34}$$

locally for $|g|\,\|z\|_\infty$ below the chart radius. The linear term vanishes by physical constrained stationarity and the gauge-zero condition. Haar density, the straightening Jacobian and the previously integrated action correction remain in the interaction; a coupling-suppressed quadratic Haar term is not asserted to vanish.

2.7 The remaining source interfaces

The chart, gauge penalty, cutoff-coordinate identity and precision comparison above concern the exact homogeneous ordinary branch already matched in Notebooks15–17. They do not extend the source's independent-free-current family: the current in (25) is the physical action derivative. Nor do they establish the full original orbit and polymer-restriction hypotheses of CMP116, or a mixed-history boundary theorem.

Finite range of Φ_V, h_V, K_V, T_V means finite dependence on the current background V. The minimizing response $W \mapsto V_*(W)$ need not have strict finite support. Weighted fluctuation-kernel locality in (27) is not a strict support-valued expansion in W or in every original fine link. A complete localized expansion of the classical remainder and the Haar, Jacobian and prior-action terms is still a separate interaction theorem. Additional source cutoff factors, when present, retain their exact pullbacks.

3 Local Haar, straightening, and exponential-gauge interactions

Use the exact current-step chart of Section 2.2. The next block side L is fixed, the current spacing is one, and constants are uniform in the preceding homogeneous depth M and compatible periodic volume. Backgrounds $V = e^{ia}$ range over a sufficiently small product ball in the independent current link coordinates $a(b)$. A slightly larger fixed ball is used for Cauchy estimates. This section localizes the measure factors; the classical and previously integrated action remainders are separate.

Write $A_{c,p} = D_{p(c)}F_V(0;c) = L^{1-d}\pi_c(V)$ and define the square ambient straightening map

$$\psi_V(B) = B - h_V\widetilde{D}_V(B), \qquad F_V(\psi_V(B)) = A_V B, \qquad \Phi_V(z) = \psi_V(K_V z). \tag{35}$$

Its retained components equal those of B. The nonlinear equation for its pivot $p(c)$ reads that pivot and no other pivot. All maps and derivatives used below are local analytic functions on fixed neighborhoods.

More explicitly, choose supports Y_r of current cells for every primitive term below. Include both its background and retained fluctuation dependencies. There are fixed constants s_L, r_L, n_L such that

$$|Y_r| \le s_L, \qquad \operatorname{diam} Y_r \le r_L, \qquad \ell(Y_r) \le s_L r_L, \qquad \sup_x \#\{r : x \in Y_r\} \le n_L. \tag{36}$$

Here ℓ is shortest-tree length; periodic multiplicities are retained. Finite enlargement of the two-block chart neighborhoods and the block-internal contour neighborhoods gives these constants. The local implicit equations do not propagate support between different coarse equations.

For nonnegative a_0, κ define the primitive local norm

$$\|l\|_{a_0,\kappa,R} = \sup_x \sum_{r: x\in Y_r} e^{a_0|Y_r|+\kappa\ell(Y_r)} \sup_{V\in\mathcal{D}_b,\, \|z|_{Y_r}\|_\infty \le R} |l_r(V,z)|. \tag{37}$$

The fluctuation supremum may be taken over complex coordinates, so it also bounds the fixed real cutoff. In particular,

$$\|l\|_{a_0,\kappa,R} \le n_L e^{a_0 s_L + \kappa s_L r_L} \sup_r \|l_r\|_\infty. \tag{38}$$

Adding primitive terms with the same support preserves this bound by the triangle inequality.

3.1 The exact ambient Jacobian and the delta normalization

Lemma 3.1 (Straightening determinant)**.** *On a fixed small complex chart,*

$$\det D\psi_V(B) = \prod_c \frac{\det A_{c,p}}{\det D_{p(c)} F_V(\psi_V(B); c)}. \tag{39}$$

Each ratio is analytic, equals one at $B = 0$, and has fixed local dependence.

Proof. Order the variables as retained coordinates followed by pivots. The retained rows of $D\psi$ are $[I, 0]$. The pivot-to-pivot block is block diagonal, because each local implicit equation reads only its own input pivot. Differentiating $F_V(\psi_V(B)) = A_V B$ in that pivot gives

$$D_{p(c)} F_V(\psi_V(B); c) D_{B(p(c))} \psi_V(B)(p(c)) = A_{c,p}.$$

Taking the determinant of this block triangular matrix proves (39). At zero, $DF_V = A_V$ and $D\psi = I$. All local blocks stay invertible on a smaller fixed domain. □

Let $\mathrm{Tr}_{\mathfrak{g}}$ denote the ordinary matrix trace in color-coordinate space. The logarithm of the normalized c factor is

$$j_c^{\mathrm{str}}(V; B) = -\,\mathrm{Tr}_{\mathfrak{g}} \log \left[I + A_{c,p}^{-1} \{ D_{p(c)} F_V(\psi_V(B); c) - A_{c,p} \} \right], \tag{40}$$

with branch zero at $B = 0$. This is a determinant identity: the diagonal Jacobian block itself is $(DF_p)^{-1} A_p$, and no commutativity of color matrices is asserted. Local analyticity and the logarithm series imply, for a fixed $\rho_L > 0$ and $|g|R < \rho_L$,

$$\| j^{\mathrm{str}}(V; K_V gz) \|_{a_0,\kappa,R} \le C_{L,a_0,\kappa} |g| R. \tag{41}$$

Indeed, the matrix perturbation inside the logarithm is at most $C_L|g|R$, which may be made at most $1/2$.

The determinant in Lemma 3.1 belongs to the square ambient change of variables, not to the rectangular derivative $D\Phi_V$. To identify its measure role, for real backgrounds and test functions supported within the chart,

$$\begin{aligned} \int dX\, \delta(F_V(X)) f(X) &= \mathcal{D}(V) \int dz\, \exp\left\{ \sum_c j_c^{\mathrm{str}}(V; K_V z) \right\} f(\Phi_V(z)), \\ \mathcal{D}(V) &= \prod_c (\det A_{c,p})^{-1}. \end{aligned} \tag{42}$$

First set $X = \psi_V(B)$ and then integrate the linear constraint $A_V B = 0$ over pivots. The pivot determinants have the positive orientation of $L^{1-d} I$ on this real neighborhood. A group delta at identity contributes only its fixed exponential-coordinate normalization at $F = 0$. After retained variables are scaled by positive real g, the additional factor $g^{\dim z}$ is field independent. Complex-background formulas are holomorphic continuations of the local densities; no complex distributional delta is used.

The factor $\mathcal{D}(V)$ is independent of fluctuations but can depend on the prescribed next-scale field through V. It is kept outside the fluctuation interaction, not discarded. Its normalized logarithm is local:

$$\log \frac{\mathcal{D}(V)}{\mathcal{D}(1)} = -\sum_c \mathrm{Tr}_{\mathfrak{g}} \log(A_{c,p} A_{1,c,p}^{-1}). \tag{43}$$

Each term is $O(\|a\|_\infty)$ and has no small coupling factor g.

3.2 The relative Haar density

Let $j(X)$ be the pullback density of Haar measure under $X \mapsto e^{iX}$, divided by its value at zero. Near zero it is nonvanishing and real analytic, with $j(0) = 1$. Haar invariance under inversion gives $j(-X) = j(X)$ on the real form and hence by analytic continuation. Its local logarithm therefore satisfies

$$\log j(0) = D \log j(0) = 0, \qquad |\log j(X)| \le C|X|^2. \tag{44}$$

This uses the positive real density and then its analytic branch; it does not complexify an absolute determinant. Translation by V preserves Haar measure, so the relative parametrization $e^{iX}V$ has the same density.

The exact Haar logarithms after scaling are

$$j_b^{\mathrm{H}}(V; gz) = \log j(\Phi_V(gz)(b)). \tag{45}$$

On retained bonds the argument is exactly $gz(b)$; on pivots it is the local chart function. From (44), the uniform local chart bound, and (38),

$$\|j^{\mathrm{H}}(V; gz)\|_{a_0,\kappa,R} \le C_{L,a_0,\kappa}|g|^2 R^2. \tag{46}$$

Any unnormalized value $j(0)$ is field independent and remains in the overall normalization. A generally nonzero quadratic Haar term remains in the interaction with coefficient g^2.

3.3 The exact retained-coordinate gauge remainder

For a root-to-site pair $r = (y, x)$ define

$$f_r(V; z) = 1 - \tfrac{1}{2}\tau \left[P_{e^{i\Phi_V(z)}V}(y, x) + P_{e^{i\Phi_V(z)}V}(y, x)^{-1} \right]. \tag{47}$$

The source contours are internal to blocks and contain no pivots. By (18), this is exactly the same expression with $\Phi_V(z)$ replaced by $E_f z$, where E_f inserts zero at pivots. Consequently it is a local function of unchanged retained coordinates.

On the full small product background domain define

$$\begin{aligned} v_r^{\mathrm{G}}(V; g, z) = -g^{-2}[&f_r(V; gz) - f_r(V; 0) - gD_z f_r(V; 0)z \\ &- \tfrac{1}{2}g^2 D_z^2 f_r(V; 0)[z, z]]. \end{aligned} \tag{48}$$

The singularity at $g = 0$ is removable, with value zero. The finite-dimensional Taylor remainder gives

$$\|v^{\mathrm{G}}(V; g, z)\|_{a_0,\kappa,R} \le C_{L,a_0,\kappa}|g|R^3, \qquad |g|R < \rho_L. \tag{49}$$

At a physical background $P_V = 1$, both $f_r(V; 0)$ and $D_z f_r(V; 0)$ vanish. Thus this is precisely the logarithmic gauge interaction after its quadratic contribution has been included in the precision. Elsewhere it is the exact third-and-higher remainder after constant and linear terms are also separated. The full-domain definition supplies background holomorphy without treating $P_V = 1$ as a product domain.

3.4 Local coefficient and multiplier bounds

Theorem 3.2 (Local measure interaction)**.** *For the fixed geometry and product background domain above, the logarithmic interactions $j^{\mathrm{str}}(V; K_V gz)$, $j^{\mathrm{H}}(V; gz)$ and $v^{\mathrm{G}}(V; g, z)$ are holomorphic and have primitive supports satisfying (36). For $|g|R < \rho_L$, their sum, regrouped by supports Y, obeys*

$$\sup_x \sum_{Y \ni x} e^{a_0|Y| + \kappa\ell(Y)} \sup |v_Y(V; g, z)| \le C_{L,a_0,\kappa}(|g|R + |g|^2 R^2 + |g|R^3). \tag{50}$$

The constants are independent of M and periodic volume. If the right side is at most one, the same bound multiplied by e holds with $|v_Y|$ replaced by $|e^{v_Y} - 1|$. For $R \geq 1$ and $|g|R \leq 1$, it is at most $C_{L,a_0,\kappa}|g|R^3$.

Proof. Combine (41), (46) and (49) and group equal supports. The local coefficient assertion can also be seen directly from the convergent expansions

$$j_c^{\mathrm{str}}(V; K_V gz) = \sum_{n\geq 1} g^n s_{c,n}(V; z),$$
$$j_b^{\mathrm{H}}(V; gz) = \sum_{n\geq 2} g^n h_{b,n}(V; z),$$
$$v_r^{\mathrm{G}}(V; g, z) = \sum_{n\geq 3} g^{n-2} v_{r,n}(V; z).$$

Each coefficient is homogeneous of degree n in z, holomorphic in only the background coordinates of its support, and has local norm at most $C_0(C_1R)^n$ by Cauchy on a fixed fluctuation disc. Background Cauchy estimates on the smaller product domain preserve the same supports. These are anchored coefficient bounds, rather than an extensive bound on the full sum. Finally $|e^v - 1| \leq e|v|$ for $|v| \leq 1$. □

The support-cardinality factor $K^{|Y|}$ in the overlapping interaction theorem is absorbed by increasing a_0 by $\log \max(1, K)$. This is a fixed-L statement, not a claim of uniform constants when L or the cell dimension is allowed to increase. For a logarithmically growing cutoff $R(g)$, these measure interactions tend to zero as soon as $|g|R(g)^3 \to 0$.

The fixed real cutoff is the original retained window $\|z(b)\| < R$. It remains unchanged because $\Phi_V(gz)(b) = gz(b)$ on retained bonds. Additional pivot, contour or reconstructed-field indicators retain their exact pullbacks and are not covered by this theorem. CMP116, p. 12 immediately before (2.3), explicitly uses the pivot-removed bond set; its (2.14), p. 15, keeps this retained cutoff fixed during Gaussian deformation. CMP109, pp. 266–268, (2.5) and (2.9)–(2.13), supplies the gauge and pivot setup. The unreadable product subscript of CMP109 (2.9) in the accessible OCR is not presented as a visually verified formula.

These factors are strictly local in independent current backgrounds V. Substituting $V_*(W)$ does not give them strict finite support in W. The full classical remainder, the previously integrated interaction, the external factor (42), and any extra cutoff indicators remain distinct obligations.

4 Localization and the next quantum induction

The homogeneous physical Gaussian is now identified in the source pivot coordinates, on a common small complex neighborhood with constants independent of the preceding classical depth. Its covariance has the spectral and spatial estimates required by the marked overlap expansion of Notebook 18. The retained cutoff coordinates are fixed, and the local measure/gauge factors have their explicit analytic bounds.

The strictly background-local version requires a small local decomposition of the actual precision and of the action remainder; exponential matrix-kernel decay alone does not supply it. Notebook 20 verifies these hypotheses directly for the bare Wilson step. Notebook 21 constructs them uniformly through the minimized classical actions using a rooted support fixed point, and closes one integration of that classical reference action. Composition with the stationary response and all inherited quantum corrections must still be included in a complete induction.

The original CMP116 localization and integration results provide one route once their orbit/reconstruction and deformed-operator hypotheses are matched. Notebook 21 supplies the direct

fixed-point route on our homogeneous small product domain. The next steps must extend to bad regions and running interactions while preserving the actual normalization.

That induction, uniform sourced observable bounds, and the continuum symmetry and reconstruction arguments remain the steps from these finite-regulator results to four-dimensional Yang–Mills existence. They are the continuing objective.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 5–18 in this workspace. The ordinary nonlinear minimum, classical coarse response and flat retained-coordinate covariance are the principal inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy, equations (0.11)–(0.16), (2.3)–(2.13).

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22, original article copy.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.